

BRYANSK, Russian FederationWMO: **268980**Lat: **53.250N**Lon: **34.317E**Elev: **216**StdP: **98.75**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.9	-18.4	-25.1	0.4	-21.3	-21.4	0.6	-17.7	9.8	-0.7	8.8	-2.3	2.0	20	0.536

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.5	29.6	19.8	27.8	18.9	26.1	18.0	20.9	27.2	19.9	25.7	19.0	24.4	3.1	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.9	23.3	17.8	13.1	22.4	16.9	12.4	21.7	61.1	27.2	57.8	26.1	54.8	24.3	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.4	7.3	6.5	-24.0	31.8	3.0	1.9	-26.2	33.2	-27.9	34.3	-29.6	35.4	-31.8	36.8		
(5)			-24.2	22.3	3.0	1.2	-26.4	23.2	-28.1	23.9	-29.8	24.6	-32.0	25.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.7	-6.1	-5.0	-0.4	8.0	14.0	17.2	19.4	18.0	12.6	6.3	0.2	-4.7		
	DBStd	10.34	5.96	5.82	4.66	4.81	4.75	3.82	3.14	3.29	4.13	4.44	5.18	5.88		
	HDD10.0	2257	498	419	324	96	17	1	0	0	21	131	296	455		
	HDD18.3	4394	756	653	582	312	148	66	24	46	174	374	546	713		
	CDD10.0	1052	0	0	1	34	142	218	292	249	101	16	1	0		
	CDD18.3	146	0	0	0	0	15	33	58	37	4	0	0	0		
	CDH23.3	1202	0	0	0	3	128	257	444	326	43	1	0	0		
	CDH26.7	278	0	0	0	0	22	52	118	84	2	0	0	0		
(14) Wind	WSAvg	2.9	3.1	3.1	3.4	3.2	2.9	2.9	2.4	2.5	2.5	2.8	3.0	3.1		
(15) Precipitation	PrecAvg	670	46	41	38	41	64	75	85	64	61	58	52	45		
	PrecMax	878	94	70	75	95	105	195	190	135	133	116	94	93		
	PrecMin	407	16	14	12	1	16	17	24	8	10	13	3	14		
	PrecStd	93	16	16	17	21	20	38	38	36	30	30	24	22		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.2	4.9	14.7	23.4	29.0	30.2	32.0	31.2	26.7	20.9	12.2	7.9		
		MCWB	3.1	2.7	8.3	13.1	17.9	19.9	20.6	20.1	16.6	14.3	10.5	7.1		
	2%	DB	2.7	3.2	10.9	20.5	26.8	28.1	29.8	29.0	24.7	17.3	9.6	4.5		
		MCWB	1.8	2.1	6.1	12.2	17.4	19.1	20.1	19.5	16.3	12.7	8.4	3.7		
	5%	DB	1.5	2.3	8.0	18.3	24.3	26.2	27.8	27.0	21.9	14.5	8.0	2.5		
		MCWB	0.9	1.3	4.0	11.0	16.3	17.9	19.5	18.4	15.3	11.3	7.0	1.9		
	10%	DB	0.6	1.5	5.8	16.1	22.2	24.5	25.8	24.9	19.2	12.4	6.5	1.4		
		MCWB	0.0	0.7	2.9	9.8	14.7	17.1	18.6	17.5	14.0	10.0	5.7	0.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.5	8.7	14.3	19.5	21.2	22.2	21.7	18.4	15.6	10.7	7.2		
		MCDB	3.9	4.1	14.2	20.8	26.4	27.9	29.1	28.4	24.5	19.0	11.6	7.6		
	2%	WB	1.8	2.4	6.3	12.7	18.1	19.9	21.1	20.5	16.9	13.2	8.8	3.7		
		MCDB	2.3	3.0	10.0	18.8	24.8	26.1	27.6	26.9	22.1	16.4	9.2	4.3		
	5%	WB	0.8	1.6	4.6	11.6	16.9	19.0	20.2	19.3	15.7	11.7	7.2	2.0		
		MCDB	1.3	2.0	6.9	17.3	23.0	24.8	25.8	25.2	20.1	14.1	7.8	2.4		
	10%	WB	0.1	0.8	3.1	10.5	15.7	17.9	19.2	18.2	14.7	10.5	5.6	1.0		
		MCDB	0.5	1.3	5.2	15.2	21.1	23.0	24.2	23.2	18.5	12.2	6.3	1.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.8	5.7	7.2	9.4	10.4	9.8	9.5	10.3	8.8	6.9	4.4	4.3		
		MCDBR	3.6	4.7	10.5	12.5	12.8	12.1	12.2	13.6	12.8	10.4	5.8	4.0		
	5% WB	MCWBR	3.5	4.3	6.5	6.7	6.0	5.4	5.3	5.9	6.3	6.6	4.9	3.8		
		MCWBR	3.3	4.2	9.5	11.3	11.8	11.0	10.5	12.1	11.0	9.3	5.1	3.9		
(40) Clear-Sky Solar Irradiance	taub		0.294	0.329	0.352	0.421	0.393	0.380	0.415	0.423	0.385	0.354	0.331	0.298		
		taud	2.135	2.064	2.122	2.137	2.313	2.370	2.291	2.252	2.327	2.376	2.257	2.159		
	Ebn at Noon		653	742	817	805	856	871	829	797	783	728	610	568		
		Edh at Noon	69	103	120	136	120	115	122	120	99	76	63	57		
(44) All-Sky Solar Radiation	RadAvg		0.75	1.71	3.11	4.07	5.50	5.78	5.51	4.71	3.08	1.54	0.62	0.45		